BHARAT HEAVY ELECTRICALS LIMITED,
TIRUCHIRAPALLI-620 014

CONTROLS AND INSTRUMENTATION/FB

PROJECT : D.B. SUPER TPP

2 x 600 MW

CUST. Nos. 1615 & 1616

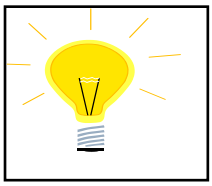
WRITE-UP ON

FURNACE SAFEGUARD SUPERVISORY SYSTEM (FSSS)

REF. DOC.NO.:- CI/1615/FSSS/WRUP

REVISION HISTORY

Rev. No.	Date	Description	Prepared	Checked	Approved
			K.Veeramani	V.M.Selvaraj	V.M.Selvaraj
00	21-06-2011	Initial Release			
01	11-10-2011	Comments received vide letter:DCPL-K8A03/DCH-4/0420			



IMPORTANT NOTE

This write up is meant for general guidelines and better understanding of the system. However, it may be noted that **"All the logic conditions"** spelt in English Language may not be exact equivalent of pictorial representation of logics. It is mandatory that the operating personal should refer to the respective contract logics.

It must be recognized that no amount of written instruction can replace intelligent thinking and reasoning on the part of boiler operators, especially when coping with unforeseen operating conditions. It is operators' responsibility to become thoroughly familiar, not only with immediate steam generating equipment but also with all pertinent control equipment. Satisfactory performance and safety depend to a great extent on proper functioning of controls and auxiliary equipment.

REFERENCE SCHEME & Dwg. Nos. FOR THE LOGIC		
SL.NO	**REFERENCE SCHEME**	**DRAWING NUMBER**
1.	Scheme Of Pulveriser With Instrumentation	1-97-291-90512
2.	Scheme Of Air And Flue Gas Path With Instrumentation.	0-97-288-90774
3.	Scheme Of Fuel Oil System	2-00-056-10345 2-00-056-10344 2-00-056-11178
4.	FSSS Logic Diagram	3-95-094-xxxxx to 3-95-094-xxxxx
5	Secondary Air Damper Control System Logic Diagram	3-95-194-24621 to 3-95-194-24628



The logic sequences are represented in English as statements, which are to be executed in certain order of precedence as described below.

The logical gates are represented in Upper case Letters, preceded and followed by "dots".

eg. .AND. , .OR.

The logical function combining two statements in the same line to be executed first.

eg. xxxxxxxxxxxxxxx .AND. xxxxxxxxxxxxxxx

The logical function in the right most position is to be executed next and so on.

eg. xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
.AND.
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

The logical function at the left most position is executed last.

eg. xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
.OR.
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

Both logical operators .AND.,.OR. have equal precedence between them.For better understanding the following example with mixed logical expressions is marked with the order of execution.

eg. xxxxxxxxxxxxx
.OR.[5]
xxxxxxxxxxxxx
.AND.[4]
xxxxxxxxxxxxxxxxxxxxxxxxx .OR.[1] xxxxxxxxxxxxxxxxxxx
.AND.[2]
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
.OR.[3]
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

NB: xxxxxxxxxxxx represent logical statements.

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15.0 Oil Elevation shutdown. (Elevation Mode)

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WINDBOX ARRANGEMENT

(SHOWN FOR CORNER – 1, SIMILAR FOR CORNERS – 2, 3 & 4

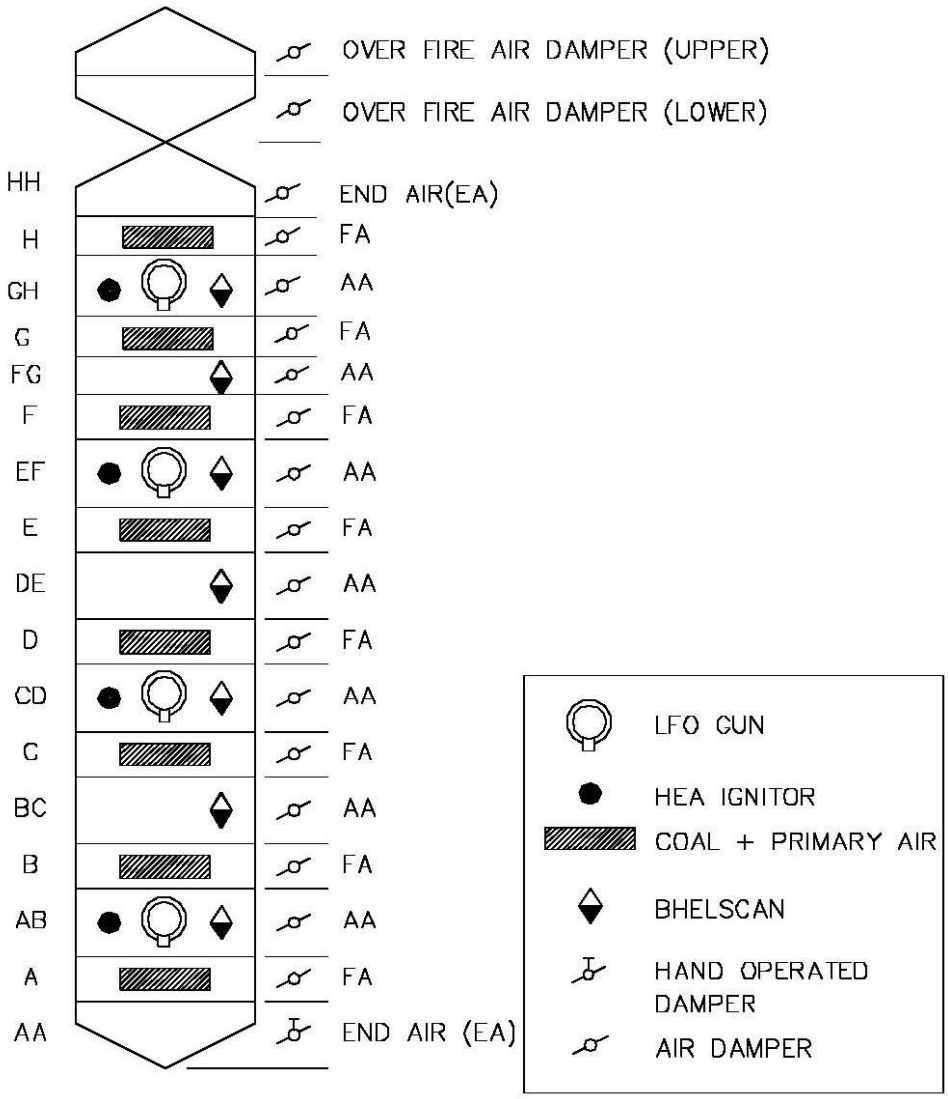


FIGURE – 1

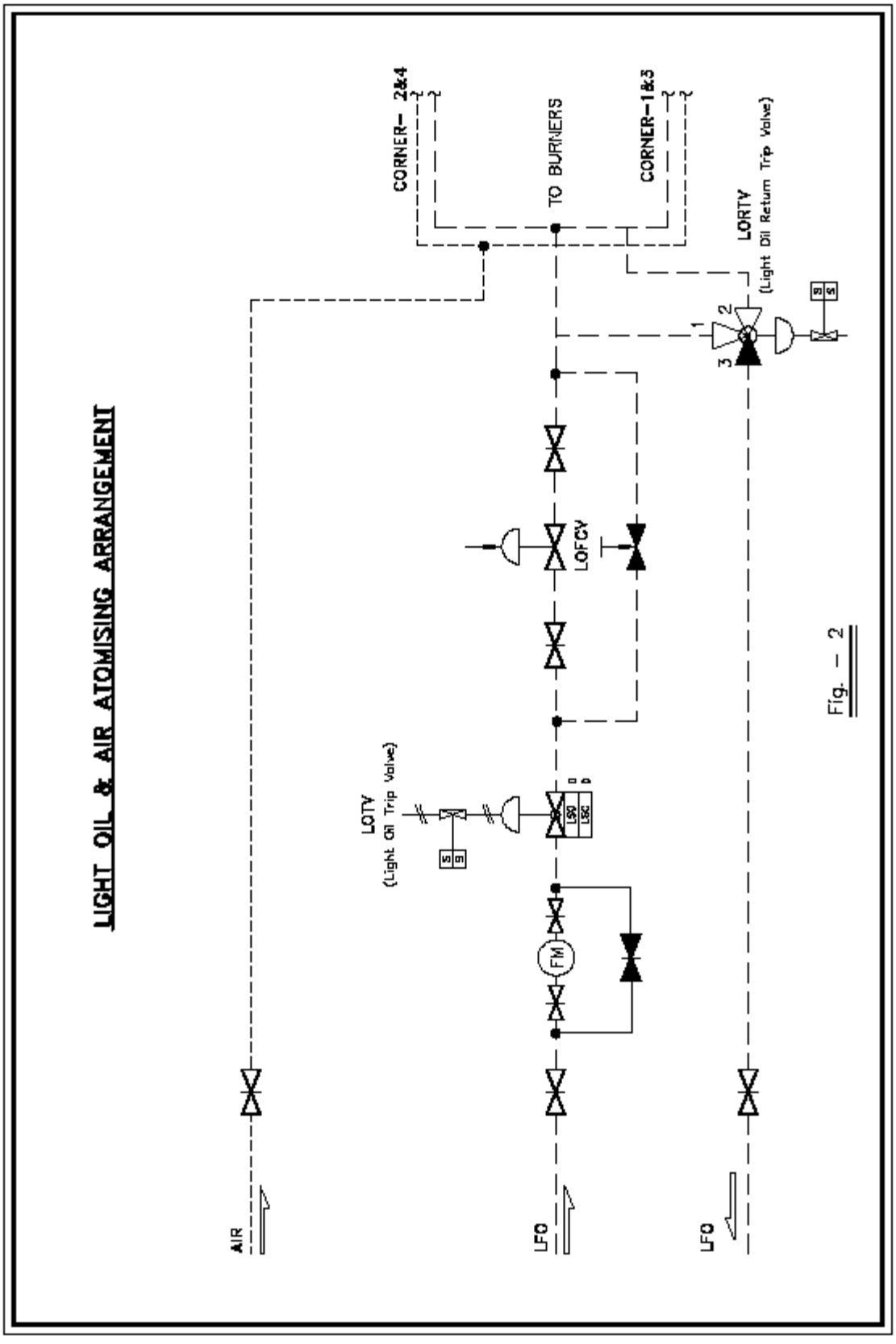

LIGHT OIL & AIR ATOMISING ARRANGEMENT
CORNER- 2&4
TO BURNERS
CORNER-1&3
LORTV
(Light Oil Return Trip Valve)
LOFCV
LOTV
(Light Oil Trip Valve)
FM
AIR
LFO
LFO

Fig. – 2

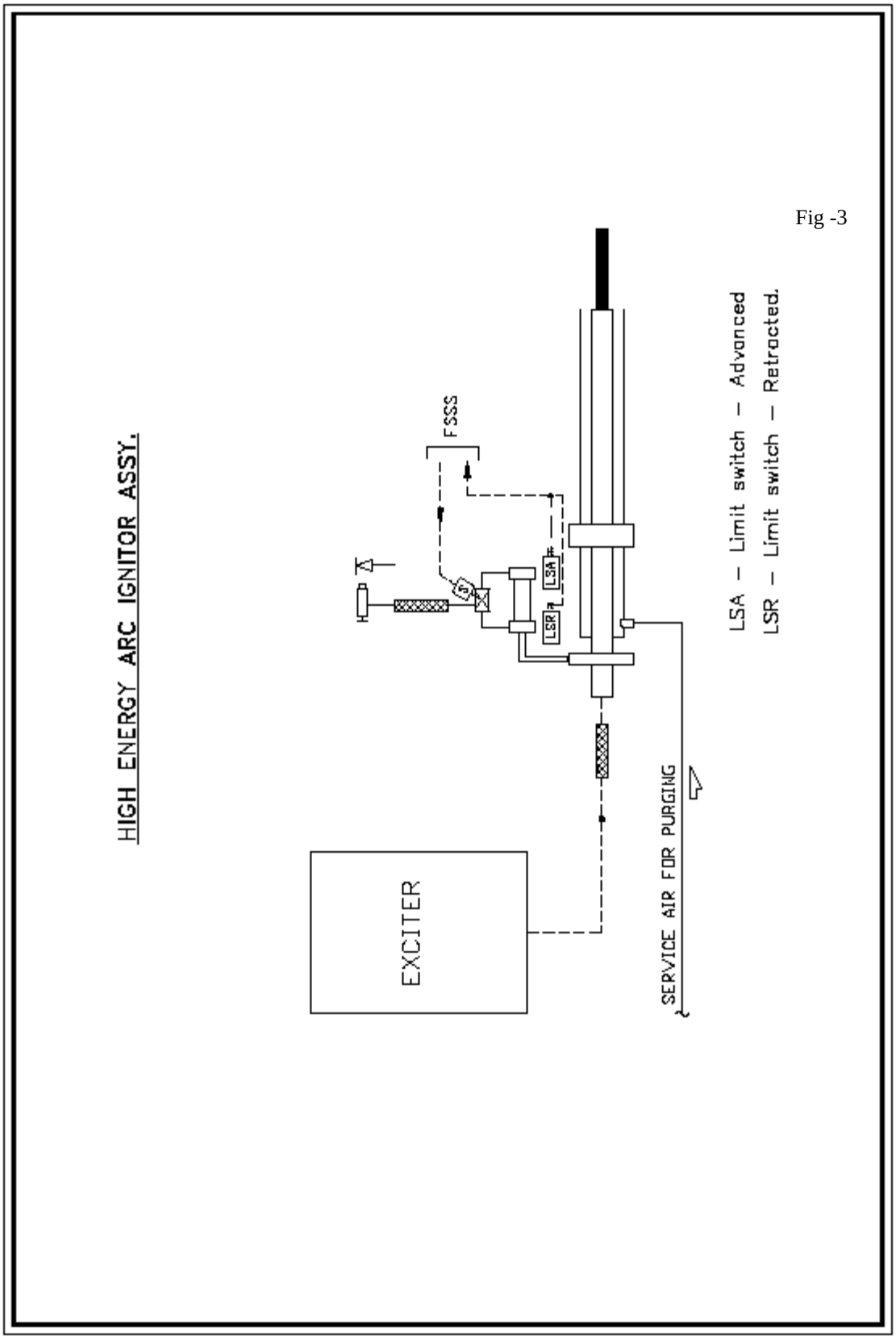

HIGH ENERGY ARC IGNITOR ASSY.
FSSS
LSA
LSR
EXCITER
SERVICE AIR FOR PURGING
LSA – Limit switch – Advanced
LSR – Limit switch – Retracted.

Fig -3

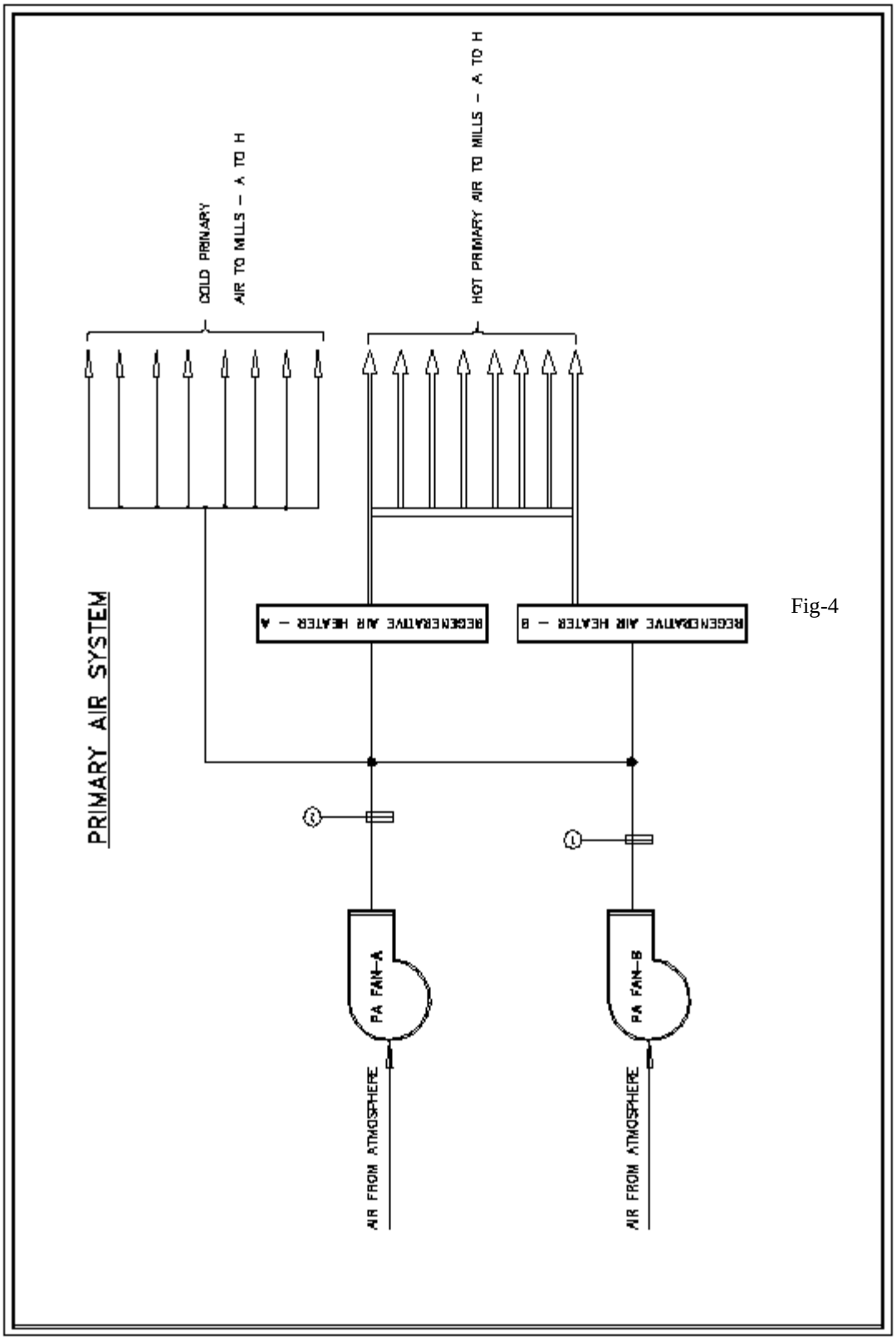

PRIMARY AIR SYSTEM
COLD PRIMARY AIR TO MILLS – A TO H
HOT PRIMARY AIR TO MILLS – A TO H
REGENERATIVE AIR HEATER – A
REGENERATIVE AIR HEATER – B
PA FAN-A
PA FAN-B
AIR FROM ATMOSPHERE
AIR FROM ATMOSPHERE

Fig-4

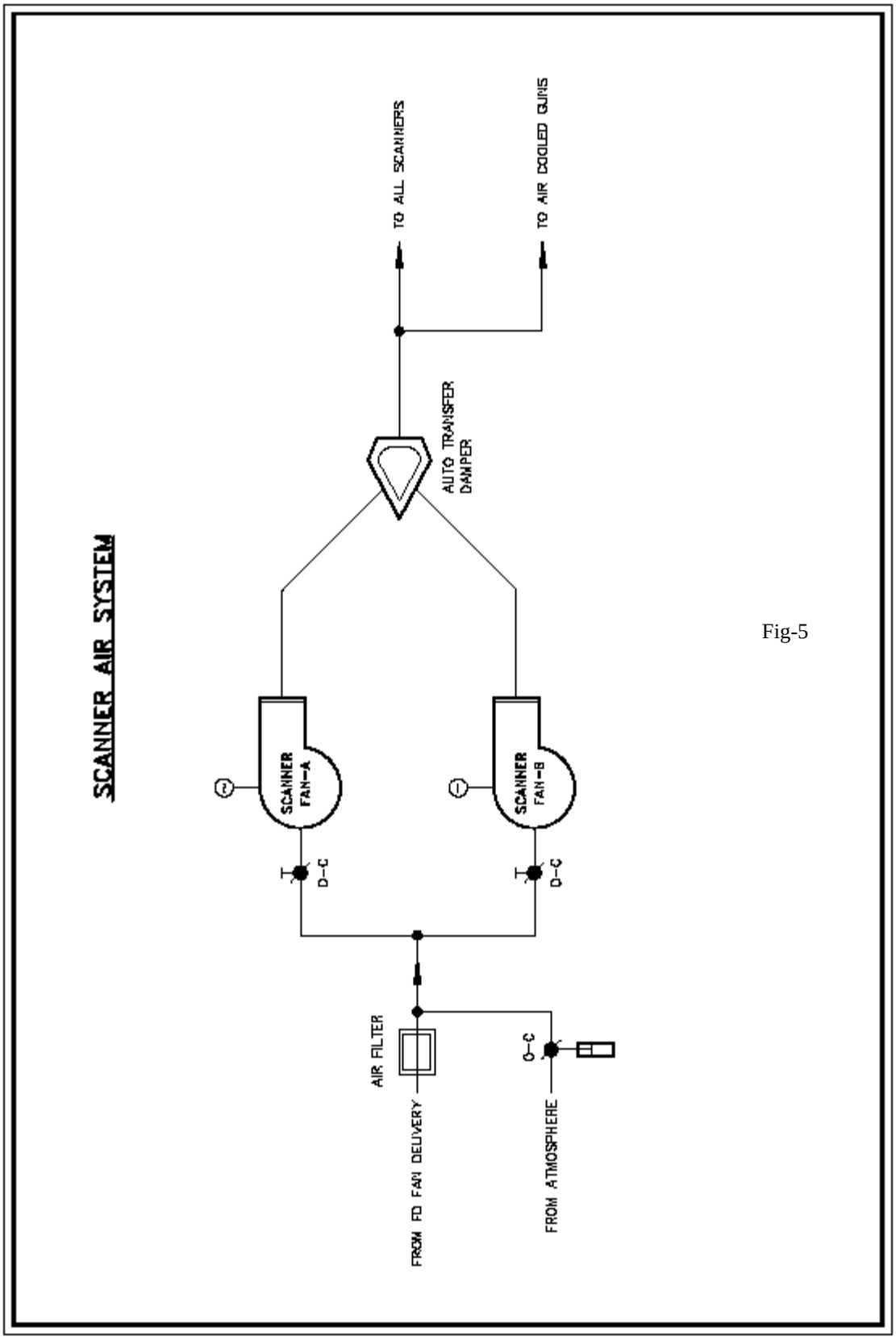

SCANNER AIR SYSTEM
TO ALL SCANNERS
TO AIR COOLED GUNS
AUTO TRANSFER DAMPER
SCANNER FAN-A
SCANNER FAN-B
D-C
D-C
AIR FILTER
D-C
FROM FD FAN DELIVERY
FROM ATMOSPHERE

Fig-5

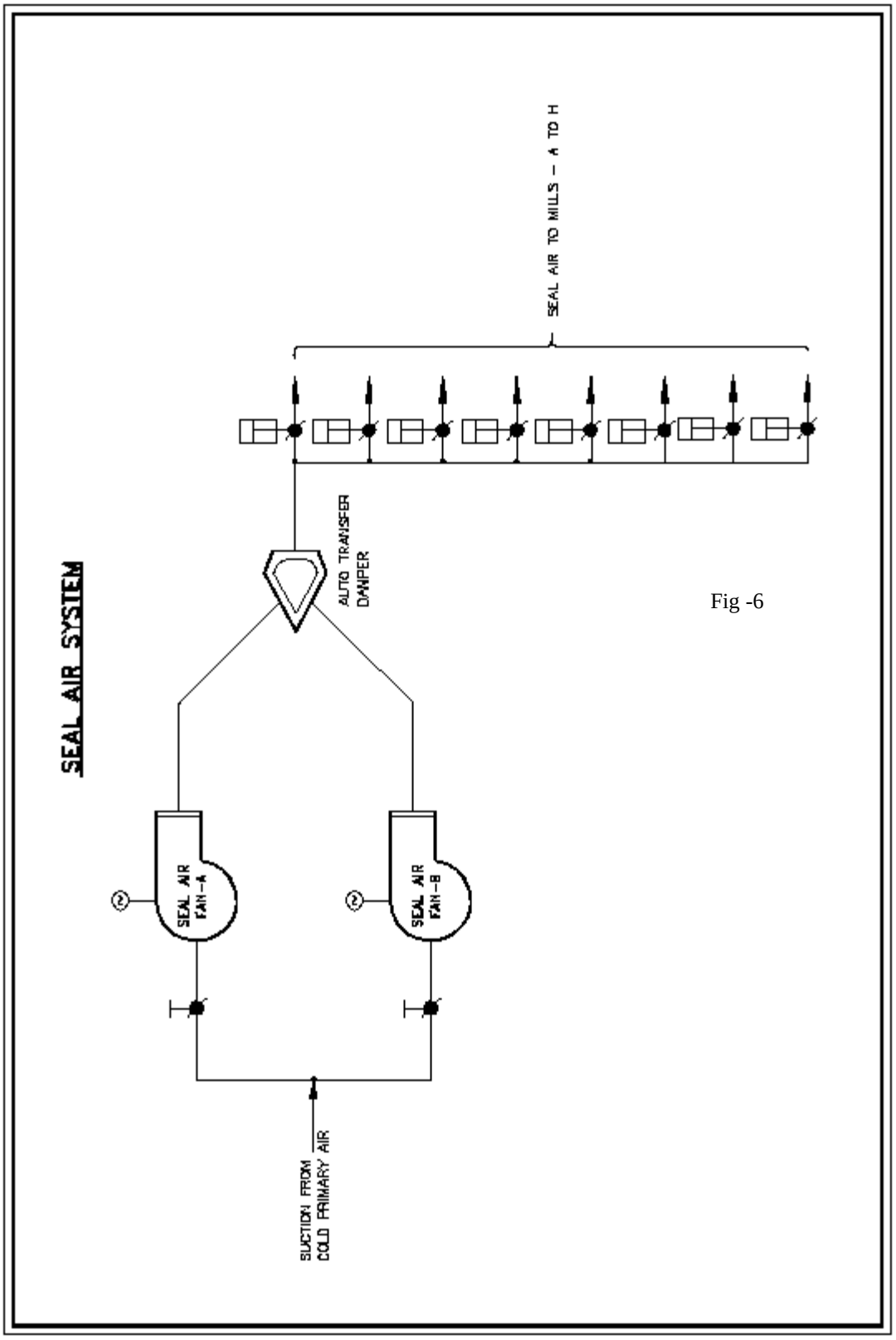
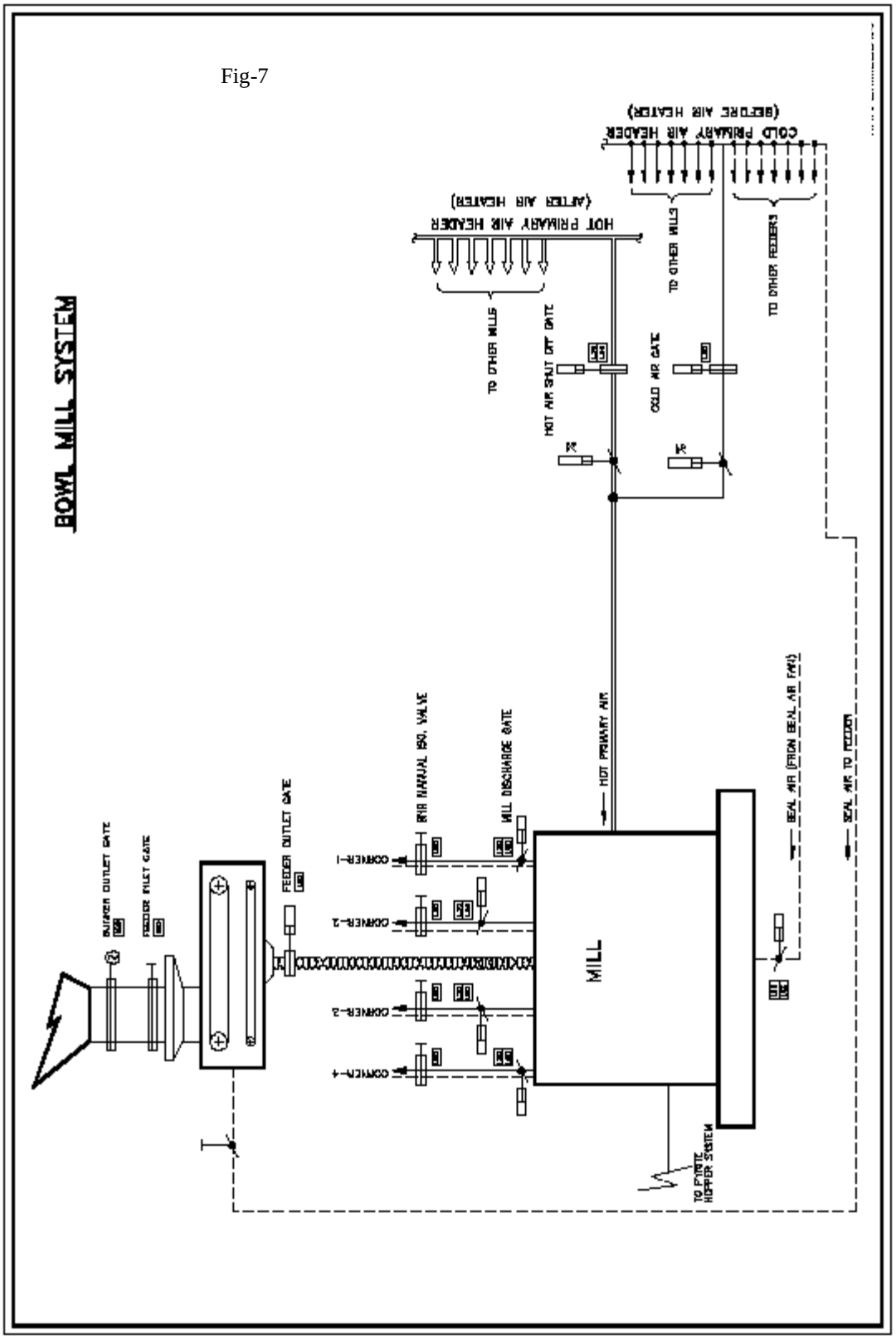

SEAL AIR SYSTEM
SEAL AIR TO MILLS – A TO H
AUTO TRANSFER DAMPER
SEAL AIR FAN-A
SEAL AIR FAN-B
SUCTION FROM COLD PRIMARY AIR

Fig -6

Fig-7


BOWL MILL SYSTEM
COLD PRIMARY AIR HEADER
(BEFORE AIR HEATER)
HOT PRIMARY AIR HEADER
(AFTER AIR HEATER)
TO OTHER MILLS
TO OTHER FEEDERS
HOT AIR SHUT OFF GATE
COLD AIR GATE
HOT PRIMARY AIR
SEAL AIR (FROM SEAL AIR FAN)
SEAL AIR TO FEEDER
BOWL DISCHARGE GATE
BOWL MANUAL ISO. VALVE
FEEDER OUTLET GATE
BURNER OUTLET GATE
FEEDER INLET GATE
CORNER-1
CORNER-2
CORNER-3
CORNER-4
MILL
TO PYRITE HOPPER SYSTEM

2.0 INTRODUCTION:-

The **Furnace Safeguard Supervisory System (FSSS)** is designed to ensure the execution of a safe, orderly operating sequence in the start-up and shutdown of fuel firing equipment and to prevent errors of omission and commission in following such a safe operating procedure.

The system provides protection, if there is a malfunction of fuel firing equipment and associated air systems. The safety features of the system are designed for protection in most common emergency situations. **However the system cannot supplement the intelligence and the reasonable judgement of the operator in all the situations.**

In some phases of operation, the Furnace Safeguard Supervisory System provides permissive interlocks only to ensure safe start-up sequence of equipment. Once the equipment is in service, the operator must use normally acceptable safe operating practices.

It is important that the operator is familiar with the overall operation of the unit and the operation of individual equipment as outlined in the various equipment sections in this manual.

It is essential that all parts of the Furnace Safeguard Supervisory System are in service at all times, if the system is to provide protection for which it is designed. **Proper maintenance and periodic inspection of the system and associated hardware is essential for continued reliable operation.**

This instruction section of the system gives a complete description of the Furnace Safeguard supervisory System furnished for this unit as it is related to the various operating phases and operation of the fuel firing equipment and associated air systems.

Basically the system is designed to perform the following functions:

- Prevent any fuel firing unless a satisfactory furnace purge sequence has first been completed.
- Prevent start-up of individual fuel firing equipment unless certain permissive interlocks have first been satisfied.
- Monitor and control the proper component sequence (Manual or Automatic) during start-up and shutdown of fuel firing equipment.
- Make continued operation of fuel firing equipment subject to certain safety interlocks remaining satisfied.
- Provide component status feedback to the operator and in some cases, to the unit control system.
- Provide flame supervision when fuel-firing equipment is in service and effect a fuel trip upon certain adverse operating conditions.
- Initiate a fuel trip when certain abnormal operating conditions exist.

Furnace Safeguard Supervisory System (FSSS) provided for this project is designed for a controlled circulation with coal as the primary fuel and Light Oil (LO) as the fuel for start-up, warm up and stabilisation. This system provides for the remote/manual operation of four elevations of High Energy Arc (HEA) ignitors, four elevations of oil guns with provision for firing Light Oil in elevations AB, CD, EF & GH and eight pulverisers serving eight elevations of 32 coal nozzles from a remote, console mounted control panel, in addition to operation through CRT (Ref. Fig.1)

From the console insert, Light Oil trip valve can be opened / closed, Purge cycle can be started, Scanner Fans & Seal Air Fans can be started / stopped, Pulveriser Discharge Gate & Hot Air Gate can be opened / closed, Pulveriser & Feeder can be started / stopped. Indicating lights on the console insert shows the operating status of various field equipments connected to FSSS.

The following operational functions are included in FSSS.

- Furnace purge supervision.
- First out trip conditions.
- Secondary air damper modulation, on / off control and supervision.
- Light oil on-off control and supervision.
- Pulveriser and feeder on-off control and supervision.
- Flame scanner intelligence and checking.
- Overall boiler flame failure protection.
- Boiler trip protection.

The Furnace Safeguard Supervisory System has been designed for **energise to open & energise to Close** philosophy. This means any on / off device or equipment has to get an impulse / command to change its present state and has to get again, another impulse / command to regain to the original state.

3.0 FURNACE PURGE:-

Before any fuel firing can take place, (initially or after a boiler trip) a successful furnace purge cycle must be completed. To start a furnace purge cycle proceed as follows:

- Establish AC & DC power to FSSS and the related equipment.
- Establish proper drum water level.
- Place atleast one ID and one FD fan in service.
- Establish air flow between 30% & 40% MCR through the furnace.

After the “Purge Ready” and “ Pust to Purge” lights come on, furnace purge cycle can be started.
When both these lights come on, it indicates that all the purge requirements are satisfied (Refer Note-1)

NOTE 1 :

If the “PURGE READY” light does not come on , check to see that:

- The light oil trip valve (LOTV) is proven closed.
- All the light oil nozzle valves are proven closed.
- All PA Fans are off.
- All feeders are off.
- All Hot air gates are fully closed.
- All the flame scanners show “No Flame” (Both fire ball and discriminating scanners).
- No boiler trip command is present (See boiler trip section for specifics).
- All aux. Air damper modulating.
- Air flow >30% and <40% MCR.

The furnace purge cycle can be started after all the purge requirements are satisfied and the “PURGE READY” light is on. The furnace purge cycle is started by depressing the START PURGE Push Button. This will start a nine minutes purge timer and the “PURGING” light will come on (see Note-2 below).

NOTE-2 :-
Loss of the purge permissive at any time during the nine minutes purging period will reset the purge timer. When this happens, the purge requirements must be re-established ("PURGE READY") and another furnace purge cycle shall be started by depressing the "PUSH TO PURGE Push Button".

At the end of the nine minutes furnace purge cycle, provided the purge permissive remain satisfied, the boiler trip is reset.

This is indicated by the "MFT" light going off, the "PURGING" and "PUSH TO PURGE" lights going off. A "NO MFT" condition is now established and the "CAUSE OF TRIP" memories are reset.

The operator may open the various trip valves as described in the following sections.

4.0 LIGHT OIL TRIP VALVE CONTROL (Ref: Fig. 2):-

Depressing the LO Trip valve "OPEN" Push Button will open the Light Oil Trip Valve if either of the following conditions are present:

LO recirculation valve is fully open
.AND.
LO Trip valve open command is present

.OR.

All Light oil nozzle valves are closed.
.AND.
The Light oil supply pressure is adequate.
.AND.
No MFT condition is present.

When the Light oil trip valve is opened the "LO Trip valve OPEN" light comes on. The "LO elevation trip command" is removed. "LO Elevation start permit" is now established.

The LIGHT OIL TRIP VALVE "CLOSE" command can be given by depressing the LO Trip valve "CLOSE" Push Button. This will close the LO Trip valve. This is indicated by "LOTV CLOSED" light coming on.

Also the LOTV closes automatically under any one of the following conditions.

Master-fuel trip .AND. LO Recirculation Valve not open

.OR.

The following conditions exist for more than 2 seconds :

Atomising air pressure very low .AND. NO LO start/stop in any Elevation.
.OR.
Light oil Header pressure very low .AND. Any Light oil nozzle valve not closed.
.OR.
Light oil trip valve not closed.

The LOTV is commanded to close on Loss of unit 24V DC.

5.0 LIGHT OIL RECIRCULATION VALVE CONTROL (Ref: Fig. 2) :-

This is accomplished by re-circulating LO through the oil re-circulation system by means of the LO trip valve (LOTV) and LO return valve (LORV).

FSSS permits the opening of LOTV for oil re-circulation purpose as follows:

Depressing the LO return valve "OPEN PB" will open the LO return valve, provided all the LO nozzle valves are fully closed.

When the LORV is fully open as indicated by the "LORV OPEN" light (Red) being "ON", depress the "LOTV OPEN PB ". This will start the LO re-circulation.

Before furnace purge cycle is started, LOTV must be proven closed. To close this, depress the associated "CLOSED PB". When the LO trip valve is fully closed, "LOTV CLOSED" (Green) light comes "ON".

To re-established the oil re-circulation after the furnace purge cycle has been completed, OPEN LOTV as described in the respective section under trip valve operation.

Depressing the "LO Recirculation valve CLOSE Push Button" will close the LO Recirculation valve.
The LO Recirculation valve will close automatically after any corner Light oil nozzle valve leaves the closed position.

6.0 LO ELEVATION START (Elevation. AB, CD, EF & GH-Pair mode)

Light oil can be fired at Elevations AB and/or CD and/or EF and/or GH either independently or in combination. Light oil elevation is placed in service on pair basis when associated Raw Coal Feeders are off. Corners 1 and 3 make one pair, while the corners 2 & 4 make the other pair. The corners making up a pair are diagonally opposite.

To place the Light oil guns in service at any elevation proceed as follows:

- Establish Light oil elevation start permit by the following conditions being satisfied.
  a) No master fuel trip.
  b) Light oil trip valve proven fully open.
  c) Pulverizer / Feeder start permit available (Ref NOTE 2)

NOTE 2 :

Pulverizer / Feeder start permit is established under following conditions:

No MFT.
.AND.
Any feeder on.
.OR.
Any oil Elevation in Service
.OR.
Nozzle Tilt horizontal .AND. Air flow < 40 %

Pressing the associated "PAIR START" push button, (for corners 1 and 3 or corners 2 and 4) when the above conditions are satisfied will initiate a 70 sec. start time to place the associated pair of Light oil guns in service as follows:

When the seventy second counting period is started, Corner-1 receives a "start" command instantaneously and corner-3 receives a "start" command after 15 seconds.

After the "start" command is established for corner-1, the corner oil gun will be placed in service if all the following conditions at corner-1 are satisfied.

- The associated corner scavenge nozzle valve is closed.
- The associated corner manual atomising steam isolation valve is open.
- The local maintenance switch in the associated corner is placed in the "REMOTE" position.

When "start" command is established at corner-1, the HEA ignitor gets advance command provided all the following conditions are satisfied.

- LO Nozzle valves are closed.
- Discriminating flame scanner shows no flame.

When the spark rod of HEA ignitor is fully advanced, the HEA Rod "ADVANCED" light comes on. Atomising valve gets open command if oil gun is engaged. Once the atomising valve is fully opened, oil nozzle valve gets open command and the exciter of HEA ignitor is energized oil nozzle valve has moved from closed position

The corner LO nozzle valve receives an "open" command provided if all of the following conditions are satisfied.

- The corner is not in scavenge.
- Atomising air nozzle valve is proven open.
- None of the corner trip condition is present and HEA Ignitor spark rod is in advanced position.
- Non existence of Corner LO nozzle valve is open and Discriminating scanner shows no flame.

When the corner oil nozzle valve (LO) is in "not closed" condition, then a 15 seconds trial time for the HEA ignitor is initiated. When the corner oil nozzle valve (LO) is proven open, the oil nozzle valve "OPEN" light comes on. At the end of the 15 seconds ignitor trial time, HEA exciter is de-energised and HEA ignitor is retracted.

If the associated corner discriminating scanner indicates flame the "DISCRIMINATING SCANNER FLAME" light comes on. Corner-1 oil gun (LO) is now in service.

When the 15 seconds counting period expires, corner-3 oil gun start command is established and will be placed in service in the same manner as previously described for corner-1.

When the seventy second counting period expires (for corner 1 & 3), the fifteen second timer for corner 3 is reset and oil gun in Corner-1 & 3 will be shut down if the associated oil nozzle valve is not fully open.

Corner No. 2 and No. 4 are placed in service by the associated "PAIR 2-4 START Push Button". The following events will occur :

• The "PAIR 2-4 START" light is illuminated and the associated "PAIR 2-4 STOP" light goes off.
• The seventy second elevation pair(for corners 2 and 4) counting period is reset.

When the Push Button is released, the 70 second counting period is started. The "Start" command for corner 2 is established and a fifteen second counting period, to establish a "Start" command for corner 4 is started. After the "Start" command is established, the associated corner oil guns (LO) are placed in service in the same manner as previously described for corner 1.

When atleast three of the four warm-up oil (LO) Nozzle valves are fully open, the warm-up oil elevation (LO) is in service. This is indicated by the "IGNITION PERMIT" light coming on the Coal Elevation Console (CRT) for the mill associated with the warm-up oil elevation. The "Ignition permit" is now satisfied and the associated mill can be placed in service after other permissives are satisfied.

When the seventy second counting period (for corners 2 & 4) expires, the fifteen seconds timer for corner 4 is reset and the oil gun in Corner-2 & 4 will be shut down if the associated oil nozzle valve is not fully open.

Light oil elevation trip occurs under any of the following conditions:

- Master fuel trip.
- Light oil trip valve not open.

7.0 ELEVATION MODE OPERATION :-

Elevation mode operation is automatically selected when either of the adjacent coal elevations associated with warm-up (LO) oil elevation is in service.

When the above condition exists, the "Elevation Mode Operation" is automatically selected and the associated warm-up (LO) oil elevation is placed in service in the following manner (provided the "warm-up elevation start permit" remains satisfied).

When pair 1-3 start or pair 2-4 start PUSH BUTTON is pressed, an "Elevation Start" command is established for corner 1 & 3. After the "Elevation Start" command is established, a forty second counting period is started to establish an "Elevation Start" command for corners 2 & 4. Corners 1 & 3 are now placed in service as reviously described in the "Pair Mode Operation" section. When the forty second counting period expires, corners 2 & 4 are placed in service as previously described in "Pair Mode Operation" section.

When atleast three of the four warm-up (LO) nozzle valves are fully open , the warm-up (LO) elevation is "proven in service".

An "Elevation Unsuccessful Start" condition is established when both seventy second counting period expires and if less than three warm-up (LO) nozzle valves are in service.

When an "Elevation Unsuccessful Start" signal is established, two seconds later, corner 1 and 3 receive "Stop" command and a forty second counting period is started. Forty seconds later corners 2 and 4 receive "Stop" commands.

An "Elevation Unsuccessful Start" condition initiates an alarm. All four corners are shut down (see "warm-up oil elevation shutdown" section for specifics).

8.0 PULVERIZER IGNITION PERMIT:-

Prior to starting any feeder, the elevation ignition energy must be adequate to support coal firing. This is accomplished as follows:

PULVERIZER"A"
A minimum of 3 of the 4 elevation "AB" LO oil nozzle valves proven open

.OR.
Boiler loading greater than 30% .AND. Feeder "B" is in service at greater than 50% loading.

PULVERIZER "B"

A minimum of 3 of the 4 elevation "AB" LO oil nozzle valves proven open.
.OR.
Boiler loading is greater than 30% .AND. Feeder "A .OR. C" is in service at greater than 50% loading.

PULVERIZER "C"

A minimum of 3 of the 4 elevation "CD" Light oil nozzle valves proven open.
.OR.
Boiler loading is greater than 30% .AND. feeder "B" .OR. "D" is in service at greater than 50% loading.
.OR.
A min. of 3 of the 4 elevation "AB" Light oil nozzle valves proven open. AND.
Feeder "B" is in service at greater than 50% loading.

PULVERIZER "D"

A minimum of 3 of the 4 elevation "CD" Light oil nozzle valves proven open.
.OR.
Boiler loading is greater than 30% .AND. feeder "C" .OR. "E" is in service at greater than 50% loading.

PULVERIZER "E"

A minimum of 3 of the 4 elevation "EF" Light oil nozzle valves proven open.
.OR.
Boiler loading is greater than 30% .AND. feeder "D" .OR. "F" is in service at greater than 50% loading.
.OR.
A min. of 3 of the 4 elevation "CD" Light oil nozzle valves proven open. AND.
Feeder "D" is in service at greater than 50% loading.

PULVERIZER "F"

A minimum of 3 of the 4 elevation "EF" Light oil nozzle valves proven open.
.OR.
Boiler loading is greater than 30% .AND. feeder "E" .OR. "G" is in service at greater than 50% loading.
.OR.

PULVERIZER "G"

A minimum of 3 of the 4 elevation "GH" Light oil nozzle valves proven open.
.OR.
Boiler loading is greater than 30% .AND. feeder "F" .OR. "H" is in service at greater than 50% loading.
.OR.
A min. of 3 of the 4 elevation "EF" Light oil nozzle valves proven open. AND.
Feeder "F" is in service at greater than 50% loading.

PULVERIZER "H"

A minimum of 3 of the 4 elevation "GH" Light oil nozzle valves proven open.

.OR.
Boiler loading is greater than 30% .AND. feeder “G” is in service at greater than 50% loading.

9.0 PULVERIZER READY

Prior to starting a pulverizer, a pulverizer start permit condition for the respective pulverizer must be established by confirming the following conditions being satisfied:

Each condition is indicated by a yellow light on FSSS console.

- Pulverizer/Feeder start permit.
- Pulverizer discharge valve open.
- Pulverizer outlet temperature not high.
- Cold air gate open .OR. Pulv. is on
- Tramp iron hopper valve open.
- Primary air permit .OR. pulverizer on.
- Pulvurizer No unsuccessful Auto start.
- All Burner isolation gates open.
- Feeder switch in Remote.
- Pulveriser no trip.
- Lube oil system healthy.
- Feeder Inlet gate open.
- Bunker Outlet gate open.

When all the above conditions are satisfied for the respective pulverizer, its associated pulverizer ready condition is satisfied.

10.0 PULVERIZER START :-

Pulverizer & the associated equipment can be started either automatically or manually. Auto / Manual mode of pulvurizer start can be selected by means of selector push buttons provided on console insert.

10.1 COLD AIR GATE OPERATION :-
CAG could be opened through OPEN PB, when Pulv. is off and once the pulv. is on. CAG is commanded to close to obtain pulv. o/l temp. effectively in a shorter duration.

Cold air gate will be commanded to open automatically when the pulveriser is on and the pulveriser outlet temperature is adequate (greater than 90° C) and will be closed automatically when the pulv. outlet temperature falls below 75° C or pulv. is off and CAG close PB is pressed.

10.2 PULVERIZER START (Manual mode):

By establishing Ignition energy available and pulverizer ready conditions for the respective pulverizer, it may be placed in service as follows:

- Ensure pulverizer ready condition and ignition energy available condition are present. Start pulverizer by pressing its associated "START" push button. First, seal air valve opens, after seal air to bowl DP adequate is established, pulverizer will start.

- When the pulverizer is proven on as indicated by the (Red) pulverizer "ON" light, open the Hot air gate by pressing its "OPEN" button and allow the pulverizer to come up to the temperature.

- When the pulverizer comes upto temperature, (approximately 120 deg. F) start the feeder by pressing its associated "START" push button. Coal flow must be proven within 5 seconds, after the feeder is started.

- Fifteen seconds after the feeder is started, the feeder output is released to the fuel totalizing circuit.

- When the feeder is on for 50 secs. the feeder speed is released to auto control and the fuel air dampers are opened to modulate as a function of feeder speed.

- When a minimum of two adjacent feeders are established at greater than 50% loading, the associated elevation of oil elevations may be shut down provided the feeder has been on for a minimum of 3 minutes.

OPERATOR's NOTE*:*

A single feeder in service for fifty seconds at its minimum speed setting will establish a "fireball" in the furnace (a single fireball is established when a coal elevation is placed in service. When additional coal elevations are placed in service, this single fireball is merely enlarged). If the support ignition energy is removed, the fireball will be monitored by the associated elevation flame scanners. However, good operator procedure dictates that an adjacent feeder be placed in service and both feeder speeds be increased to 50% before removing the support of the fireball. When removing feeders from service the inverse procedure should be followed and support ignition energy should be reinstated when only two feeders remain in service at 50% feeder speed.

10.3 PULVERIZER START (Auto mode):

The selected pulverizer can be placed in operation automatically if all the following conditions are satisfied:

- The associated " Pulverizer Ready " condition remains satisfied.
- The " AUTO " mode is selected for the associated pulverizer.

The pulverizer is started by momentarily depressing the pulverizer "START" push button. The following events will occur;

a) A 180 second counting period is started to prove the associated pulverizer / feeder in service.

b) If the pulverizer ignition energy permissive is not satisfied an "auto start support ignition " command is established and the associated LO elevation is automatically placed in operation. (Refer to " Elevation Mode Operation" section for details).

c) When the associated support ignition elevation start time has expired (both seventy second " pair start " counting periods have expired) and the pulverizer ignition permissive is satisfied, the " auto start pulverizer " and "open Hot air gate" commands are established simultaneously.

d) A pulverizer " Auto start " command is established and the the pulverizer is started in the same manner as previously described in pulvurizer manual mode.

e) When the pulverizer is proven ON the following events occur:

- The Hot air gate is opened, provided the pulverizer outlet temperature is less than high value (>100"C)

- A sixty second counting period is started to establish an "auto start feeder" command.

f) When the sixty second counting period expires, an "auto start feeder" command is established for five seconds. A "start feeder " command is now established, provided the Hot air gate is open. The feeder is put into service in the same manner as previously described. After the 180 second "elevation start" counting period expires and the pulverizer is ON and the feeder is OFF, the unsuccessful elevation start is established. When the feeder is OFF for more than three minutes, the pulverizer receives an "auto stop pulverizer" command and is shutdown.

11.0 PULVERIZER IN SERVICE

With the pulverizer / feeder in service, several operational parameters are continuously monitored. However it is left to the operator's discretion to investigate and take further corrective action.

An operating pulverizer and feeder are subject to the following operational checks:

- The feeder speed will be run back to minimum if the pulverizer bowl differential pressure is high .OR. if the respective Pulveriser Motor current is High.
- If the pulverizer outlet temperature is high (greater than 100 Degree C), then the following events occur:

    - Pulverizer air and temperature control is released from auto mode. The hot air dampers close and the cold air dampers open to 100% pulverizer air flow position.

    - The hot air gate is closed after a thirty second time delay to allow the associated dampers to react.

Any of the following conditions will give signal to run the feeder speed to minimum until the initiating condition is corrected.

- Pulverizer bowl differential pressure high.

- No Feeder start.

Loss of coal flow / coal on belt and low pulverizer current will trip the feeder and close hot air damper. The operator should take appropriate corrective action to establish coal flow to feeder belt, then reopen the hot air damper and restart the feeder.

12.0 PULVERIZER TRIP SIGNALS:-

The following conditions will initiate a pulverizer trip command.

- Loss of PA fan
- MFT (See boiler trip section for details)
- Pulverizer discharge valve not open
- Pulverizer ignition permit is not available .AND. support ignition required
- Seal air header to pulverizer differential pressure low (125 mm WC) .AND. Pulvurizer on for more than 60 secs.
- Any coal burner of that elevation Manual Isolation Valve not open.
- Loss of primary air.
- Loss of unit critical power.
- Pulverizer failed to start >10 secs.
- Pulverizer Lub oil system failed.

- Loss of unit +24V DC power.
- Mill Motor Electric Trip.

When the primary air duct pressure falls below the low set point (for more than 5 secs.) all pulverizers in service receive stop command.

When both PA fans stop or the primary air duct pressure goes very low, all the pulverizers are tripped instantaneously.

When one PA fan or one FD fan trips and four or more pulverizers are in service, each pulverizer with a 2 sec time delay a trip command will be given starting from the pulverizer serving top most coal elevation, until the number of pulverizers remaining in service is reduced to three.

13.0 PULVERIZER SHUT DOWN:-

13.1 PULVERIZER SHUT DOWN (Manual mode):

When manual pulverizer control mode is selected, manual shut down of a pulverizer is accomplished as follows:

- Start the associated oil elevation, if it is not in service.
- Reduce the feeder speed to minimum.
- Close the Hot air gate by depressing its "CLOSE" push button.
- When the pulverizer outlet temperature is reduced to below 49.5 deg C then shut down the feeder by pressing its "OFF" push button.
- Allow the pulverizer to run for about 2 minutes to ensure that it is completely empty of coal, then shutdown the pulverizer by pressing its "OFF" push button.

13.2 PULVERIZER SHUTDOWN (Auto mode):

When in the auto mode of operation the pulverizer is shutdown as described below.

a) The operator initiates a pulverizer auto shutdown sequence by momentarily depressing the associated "STOP" push button. The following events will occur.

i) The "auto stop memory" signal is established and the "auto start memory" is reset.

ii) If the associated pulverizer ignition permissive is not satisfied, an "auto start support ignition energy" command is established.

iii) A ten minutes "elevation stop" counting period is started.

b) When the support ignition time has expired (both seventy second LO pair starting periods have expired) and the pulveriser ignition permissive is satisfied, then the following commands are established simultaneously.

i) Run the feeder speed demand to minimum. The feeder speed will be reduced through auto control.

ii) Auto close Hot air gate. The associated Hot air gate is closed in the manner previously described (see PULVERISER MANUAL SHUT DOWN section).

c) When the Hot air gate is closed and the pulveriser outlet temperature has been reduced below 49 Degree C (120 deg.F) the associated feeder will receive an "auto stop" feeder command. The feeder will now be shut down.

d) When the feeder is off for three minutes, the associated pulveriser will receive an "auto stop" pulveriser command. The pulveriser will now be shut down.

e) If the pulveriser is to remain on "stand by" then the discharge valve should remain open to allow air to flow through the pulveriser. If required , the discharge valve is closed by depressing the associated close push button (provided the pulveriser is off).

The associated LO elevation can be shutdown when the furnace and boiler conditions have stabilised.

When the ten minutes "elevation stop" count period expires, an "unsuccessful elevation shut down" signal will be sent to the alarm if any of the following conditions exist at the associated coal elevation:

i) The pulveriser is on.
ii) The feeder is on.
iii) The Hot air gate is not closed.

13.3 FEEDER OUTLET GATE CONTROL :-

Depressing “OPEN” Push Button for RC Feeder Outlet Gate will open the gate provided the following conditions are fulfilled.

- Pulveriser on.
- Close RC Feeder Outlet Gate command is not present.
- RC Feeder Outlet Gate not open.

Start Feeder - Auto command will automatically open the gate.

“Red” lamp will indicate open state of the gate.

Similarly by depressing “CLOSE” Push Button for RC Feeder Outlet Gate will close the gate provided RC Feeder Outlet Gate is not closed and the following conditions are fulfilled.

Feeder off .AND. RC Feeder Outlet Gate close command present.

.OR.

Pulveriser trip

.OR.

Feeder off .AND. Stop feeder - auto.

“Green” lamp will indicate close state of the gate.

13.4 HOT AIR GATE CONTROL :-

Pressing Hot Air Gate "OPEN" Push Button on, will open the gate if following conditions are present.

Pulveriser on

.AND.

Pulveriser auto start

.OR.

Pulveriser manual mode

.AND.

Close Hot air gate command is not available.

"Red" lamp will indicate open state of the gate.

Hot Air Gate will close automatically if following conditions are present.

Close Hot Air Gate command present for more than 30 seconds.

.OR.

MFT

.OR.

pulveriser off

Loss of unit 24V DC will also close the Hot Air Gate.

This is indicated by "Green" lamp.

14.0 OIL ELEVATION SHUTDOWN:-

The oil elevation can be shutdown in either "PAIR" mode or "ELEVATION" mode manually. When all feeders are off .AND. associated Pulverizer of the oil elevation is in manual mode then " PAIR " mode is selected. If either feeder associated with the oil elevation is proven ON the "ELEVATION" mode is selected.

15.0 PAIR MODE SHUTDOWN (for elevation AB ,CD , EF & GH):-

When adjacent coal elevations are not in service and elevation back up trip exists or when all coal elevations are off, the associated warm-up (LO) elevations are automatically selected to the "Pair operation Mode" and the warm-up (LO) elevation can be shut down, pair-wise.

Under "pair operation mode" the oil elevation is removed from service in the following manner.

Either pair (1-3) or (2-4) can be initially shutdown. Each pair of oil guns are shutdown in the following manners.

Pressing "PAIR (1-3 or 2-4) STOP" Push Button momentarily will establish a stop signal for that pair. This will cause the following events to occur :

A 375 second elevation "Pair Stop" counting period is started.

The associated Elevation "PAIR (1-3 or 2-4) STOP" light comes on.

A “Stop” command for corner 1 (or corner 2) is established, and a fifteen seconds counting period is started to establish a “Stop” command for corner 3 (or corner 4).

The corner oil (LO) nozzle valve will receive a “CLOSE” command.

(When the corner oil nozzle valve is moved to the fully closed position, the oil valve “CLOSED” light comes ON and the associated “DISCRIMINATING SCANNERS NO FLAME” light comes on.)

After the warm-up (LO) nozzle valve is closed, the scavenge nozzle valve receives “Open” command. Also the spark rod of HEA is advanced and gives spark for a period of 15 seconds. After the scavenge valve is fully open, a five minute counting period is started.

After the five minutes scavenge time expires, the Scavenge valve receives “Close” command. When the Scavenge valve is moved from the fully open position, the five minute counting period is reset i.e. scavenge command is reset.

When the Scavenge command is reset, the associated atomising (Air) nozzle valve receives a “Close” command.

When the fifteen second counting period expires, a “Stop” command is established for corner 3 (or corner 4) which is removed from service in the same manner as previously described for corner 1 (or corner 2).

After the pair stop time expires, the scavenge valve memory is reset.

The other pair (1-3 or 2-4) is stopped similarly by depressing the “PAIR STOP” Push Button. (see Note-4 below).

NOTE4:

Simultaneously, a 360 second counting period is started to establish a “warm-up” elevation back up trip after the first pair (1-3 or 2-4) of oil guns have been shut down and the “Stop” command for the first oil gun of the second pair (1-3 or 2-4) is established.

The associated pair of oil guns are shut down in the same manner as previously described for the other pair of oil guns (1-3 or 2-4).

When the 360 second counting period expires, a warm-up “Elevation Back up trip” condition is established. This pair “Back up trip” establishes a redundant “Stop” command to ensure that the oil guns are shut down.

The signal to “Position the warm-up oil elevation dampers for warm-up fuel firing” is removed. After the “warm-up oil elevation back up trip” is established, an “Unsuccessful warm-up oil elevation shut down” condition is established when both 375 second counting period expire and any warm-up (LO) nozzle valve is not closed or any HEA Ignitor is not retracted in the associated elevation. This initiates “Oil Elevation Unsuccessful Shut down” alarm.

A warm-up oil (LO) gun may also be shut down and scavenged individually by placing its local gun maintenance switch in the “SCAVENGE” position. The warm-up oil gun will be shut down and scavenged as described above.

To return the warm-up oil gun to service, return the local gun maintenance switch to “REMOTE” position and depress the associated warm-up oil elevation “PAIR START” Push Button.

Any one of the conditions listed below will trip an individual warm-up oil corner by issuing a corner trip signal.
• Associated local maintenance control switch turned to “OFF” position.
• Warm-up oil gun not engaged.
• Atomising (Air) Manual isolating valve not open.

Corner not in scavenge
.AND.
Corner start time expired
.AND.
Warm-up oil nozzle valve not open
.OR.
Atomising air nozzle valve not open
.OR.
Discriminating scanner no flame
Elevation back up trip.

15.1 OIL ELEVATION SHUTDOWN (ELEVATION MODE) :-

When either of the coal elevation associated with the warm-up oil elevation, is in service, the warm up oil elevation can be shut down under this elevation mode. The subject elevation can be shut down as described in the following section.

Under Elevation mode, the oil elevation is removed from service in the following manner.

Pressing the “PAIR STOP” Push Button (1-3 .OR. 2-4) will establish a corner “STOP” command (see NOTE4 below).

NOTE4 :-
The “PAIR STOP” command will establish a corner “STOP” command provided both coal elevations associated with the warm-up elevation have been ON for at least three minutes or both associated coal elevations are not in service.

The corner “STOP” command will cause the following events to occur:

- A 375 second elevation pair counting period is started for corner 1 and 3.
- A 40 second counting period is started to establish a “STOP” command for corner 2 and 4.

The Elevation “PAIR 1-3 STOP” light comes on. A “STOP” command for corner 1 is established and a 15 second counting period is started to establish a “STOP” command for corner 3.

After the “STOP” command is established, the oil gun at corner 1 is removed from service in the same manner as previously described.

When the 15 second counting period expires, the “STOP” command for corner 3 is established and is removed from service in the same manner as corner 1. When the 40 second counting period expires, the “STOP” command is established causing the following events to occur. A 375 second elevation pair counting period is started for corners 2 and 4. The Elevation “PAIR 2-4 STOP” light comes on.

Corner 2 stop command is established and a 15 second counting period is started to establish a stop command for corner 4.

The oil gun at corner 2 is removed from service in the same manner as previously described.

When 15 second counting period expires the "STOP" command for corner 4 is established and is removed from service in the same manner as corner 2. All other interlocks i.e., Backup trip, elevation unsuccessful shutdown etc. are same as described earlier.

16.0 BOILER TRIP CONDITIONS :-

16.1 MASTER FUEL TRIP (MFT)

A MASTER FUEL TRIP command stops all fuel inputs by tripping all the pulverizers, all the feeders and all oil guns, closing the oil trip valve and closing the light oil trip valve.

Master Fuel Trip is initiated by any one of the following conditions :
- Less than fire ball and loss of AC at any elevation in service (See the NOTE-5).
- Drum water level very low for more than 5 seconds.
- Drum water level very high for more than 10 seconds.
- Both FD Fans off.
- Both ID Fans off.
- Air flow is less than 30%
- Furnace pressure very low.
- Furnace pressure very high.
- Loss of all fuel trip (See the NOTE-6)
- Unit flame failure trip (See the NOTE-7)
- Both emergency trip Push Buttons pressed (Operator initiated boiler trip)
- Spare trip.
- Loss of Reheater protection.
- Loss of 220V DC for more than 2 second.
- Boiler feed pumps tripped.
- Waterwall circulation not correct.

NOTE-5 :-
"Less than fire ball and loss of AC in any elevation in service" condition occurs, when all the mills are not in service and there is a loss of power supply (110 V AC-for Solenoids) at any of the elevation in service. Under this condition as ignition energy will not be proper (no fire ball), the LO elevation (Oil) cannot be allowed to run without control (as there is no power supply). Hence the boiler has to be tripped to shut down the elevation.

NOTE–6 :-
"Loss of All Fuel Trip" circuit is armed as soon as oil firing is introduced into the furnace, through any one of the following conditions. i.e. if any oil burner comes into service. This condition also indicates that the "Boiler is on".

This condition is generated when any one of the following condition occurs :
Elevation AB any LO nozzle valve .NOT. closed
Elevation CD any LO nozzle valve .NOT. closed
Elevation EF any LO nozzle valve .NOT. closed
Elevation GH any LO nozzle valve .NOT. closed

16.2 LOSS OF ALL FUEL TRIP :
"Loss of All Fuel Trip" condition is generated if all the following conditions existalong with "Loss of all fuel trip arming" condition.

All feeders are off
.AND.
Elevation AB back up trip or All LO valves are closed
.AND.
Elevation CD back up trip or All LO valves are closed
.AND.
Elevation EF back up trip or All LO valves are closed
.AND.
Elevation GH back up trip or All LO valves are closed
.OR.
LO Trip Valve not open.
Loss of all fuel trip arming memory is reset 5 seconds after MFR trip occurs.

16.3 UNIT FLAME FAILURE :-
"Unit Flame Failure Trip" occurs if all the 8 elevations vote for No flame along with the arming signal. Conditions for the generation of No Flame voting Elevation wise is given below :

ELEVATION-A
Feeder-A is not proven.
.OR.
Elevation-AB 3 out of 4 Fireball scanners show no flame .AND. Elevation-AB 2 out of 4 nozzle valves not open.
.OR.
Elevation-AB started and loss of power.

ELEVATION-B
Feeder-B is not proven
.OR.
Elevation-AB 3 out of 4 Fireball scanners show no flame .AND. Elevation-AB 2 out of 4 nozzle valves not open .
.OR.
Elevation-AB started and loss of power.
.AND.
Elevation-BC 3 out of 4 fire ball scanners show no Flame

ELEVATION-C
Feeder-C not proven.
.OR.
Elevation-CD 3 out of 4 Fireball scanners show no flame .AND. Elevation-CD 2 out of 4 LO nozzle valves not open .
.OR.
Elevation-CD started and loss of power.
.AND.
Elevation-BC 3 out of 4 fire ball scanners show no flame

ELEVATION-D
Feeder-D not proven.
.OR.
Elevation-CD 3 out of 4 Fireball scanners show no flame .AND. Elevation-CD 2 out of 4 LO nozzle valves not open .

.OR.
Elevation-CD started and loss of power.
.AND.
Elevation-DE 3 out of 4 fire ball scanners show no flame

ELEVATION-E
Feeder-E not proven.
.OR.
Elevation-EF 3 out of 4 Fireball scanners show no flame .AND. Elevation-EF 2 out of 4 LO nozzle valves not open .
.OR.
Elevation-EF started and loss of power.
.AND.
Elevation-DE 3 out of 4 fire ball scanners show no flame

ELEVATION-F
Feeder-F not proven.
.OR.
Elevation-EF 3 out of 4 Fireball scanners show no flame .AND. Elevation-EF 2 out of 4 LO nozzle valves not open .
.OR.
Elevation-EF started and loss of power.
.AND.
Elevation-FG 3 out of 4 fire ball scanners show no flame

ELEVATION-G
Feeder-G not proven.
.OR.
Elevation-GH 3 out of 4 Fireball scanners show no flame .AND. Elevation-GH 2 out of 4 LO nozzle valves not open .
.OR.
Elevation-GH started and loss of power.
.AND.
Elevation-FG 3 out of 4 fire ball scanners show no flame

ELEVATION-H
Feeder-H not proven.
.OR.
Elevation-GH 3 out of 4 Fireball scanners show no flame .AND. Elevation-GH 2 out of 4 LO nozzle valves not open .
.OR.
Elevation-GH started and loss of power.
.AND.
Elevation-FG 3 out of 4 fire ball scanners show no flame

NOTE-7 :-
"Unit Flame failure Trip" circuit is armed 2 seconds after any coal elevation is put into service.

16.4 FIRST OUT TRIP SYSTEM :-
A First Out Trip System provided allows the operator to determine the cause of aboiler trip by observing the indication lamps located on the Unit Console insert underFirst Cause of Trip.

When an event that causes occurrence of boiler trip, the associated “Cause Of Trip”memory is set, and the indicating lamp for trip command, responsible for the boiler trip is illuminated and the successive trip commands to the other indications are blocked.

When the event that caused the boiler trip is corrected & boiler purging is completedthe “Cause Of Trip” memory is reset.
This indicates that a “No Boiler Trip Command” is established. The memories are reset automatically after a successful 9 minute furnace purge cycle has been completed. The FIRST CAUSE OF TRIP light that caused the boiler trip will go off.

16.5 POST PURGE EXCURSION PROTECTION :-

Post purge excursion protection circuit has been designed in the Furnace Safeguard Supervisory System. It operates as described below:
When all the following conditions are satisfied, a 5-minute counting period is started.
• Light oil trip valve closed.
• All elevation all LO nozzle valves closed.
• All Feeders off.
• Air flow > 30 % MCR and <40 % MCR.
• All scanners show “no flame”.
• All Hot air gates closed.
• All aux. air dampers modulating
• All PA fans off.
After 5 minute counting period expires, the “Post Purge Fan Trip” memory is set, if very high or very low furnace pressure exists, the post purge fan trip command is issued (for tripping FD&ID fans). When any LO valve is open, this is reset.

16.6 FEED FORWARD COMMAND TO FURNACE DRAFT CONTROL :-

A 30 seconds Feed Forward command to Furnace draft control is given if either of the conditions listed below exists.
• Loss of unit +24V DC power
• Occurrence of Master Fuel Trip, when the boiler is on (i.e., Loss of all fuel trip
arming is present).

16.7 FURNACE PRESSURE RECORDER SPEED CHANGE :-
This contact has been provided to switch the Furnace Pressure Recorder to the alternate speed (if Dual speed recorder is provided) under the conditions mentioned below, to observe Furnace Draft fluctuations.
• MFT
• Loss of unit +24V DC power
• Furnace Pressure High or Low
For condition 1 & 2 mentioned above, the Recorder speed is reverted back to original value after 30 seconds. For condition 3, the Recorder is set to original value 30 seconds after condition 3 is removed.

17.0 SCANNER AIR FAN CONTROL:

Scanner fan “A” or “B” can be started manually by depressing the respective “START PB”.
Scanner fan “A” or “B” will start automatically whenever the scanner duct to furnace differential pressure goes below 150 mm W.C. .AND. No MFT .OR. When there is loss of unit +24 V DC.
If one scanner fan fails to start within 5 secs. after receiving the start command, the other scanner fan will start automatically.

Scanner fans A or B can be stopped by depressing the respective "STOP PB" provided there is No start command .AND. Scanner duct to furnace differential pressure is more than 150mm W.C.
When both the FD fans are off .OR. Loss of unit +24 V DC, scanner emergency damper will open automatically. Damper closes automatically when any FD fan is ON.

18.0 SEAL AIR FAN CONTROL:-

There are two seal air fans that serve this unit, one working and one standby.

Any seal air fan can be started manually by depressing its "START PB". This action gives start command to the motor.
The seal air fan which is not in service, is started on Auto command if, Any PA fan is on .AND. Seal air header to cold PA header DP is low for > 10 secs.
The seal air fan "A" will start automatically 5 second after the seal air fan "B" trips or fails to start on receiving start command (SA fan "B" starts automatically if SA fan "A" fails to start).
Any seal air fan can be stopped by depressing its "STOP PB", provided the following conditions are satisfied.
Seal Air Header / Cold Air duct DP not low
.AND.
Both Seal Air fans on.
Both Seal Air Fans are tripped once Both PA Fans are off > 15 Secs.

19.0 SECONDARY AIR DAMPER CONTROL:-

For a complete description of the Secondary Air Damper Control, refer the write up on SADC system. Operation of the secondary air damper control as effected by the FSSS is described here.

19.1 AUXILIARY AIR DAMPER CONTROL

All elevations of the aux. air dampers will be automatically switched to "Manual" control and then commanded to open 100%, if any one of the following conditions exist.
• All FD fans are off
• All ID fans are off
• Master fuel trip exists (for more than 2 seconds PULSE)
• Loss of Unit +24 V DC power
When the auxiliary air control is placed in the "Auto" mode of operation, the auxiliary air dampers modulate to control wind box to furnace differential pressure at a predetermined set point.

During the furnace purge cycle, all auxiliary air dampers modulate to maintain wind box to furnace differential pressure at a predetermined set point.

When the unit load exceeds 30% of MCR, the following events occurs : The wind box to furnace differential set point is gradually increased as the unit load is increased. Conversely, the differential pressure set point is gradually decreased as unit load is decreased. The rate of increase (or decrease) is controlled by the auxiliary damper controller.

When the Unit Load is increased above 30% MCR, the Auxiliary air dampers on elevation AB, BC, CD, DE, EF, FG, GH & HH are closed in a timed sequence (10 seconds interval) provided the associated coal elevations are not in service and the associated oil elevations are not in service. The auxiliary air dampers are closed, starting from the top elevation, progressing towards the lowest elevation.

When the unit load is reduced below 30% of MCR, the following events occur:
• The wind-box to furnace differential pressure set point reverts to its original value.

• The auxiliary air dampers associated with the coal and LO elevations not in service are opened in a timed sequence (10 second interval) starting at the lowest elevation, progressing towards the top.

When a LO Elevation is started (i.e. when the associated “elevation back-up” trip is removed) the associated auxiliary air dampers in that elevation are set to preset position for LO firing.

When the “Post Trip Purge Time Expired” command is established and all FD / ID fans are not off and a Master fuel trip” command doesn’t exist, then a 35 seconds counting period is started. After it expires, the “Open All Auxiliary Air Dampers” command is removed from memory.

19.2 FUEL (COAL) AIR DAMPERS CONTROL :-

All elevations of fuel air dampers are commanded to open 100% when any one of the following conditions exist.

• All FD fans are off
• All ID fans are off
• Master fuel trip exists (for more than 2 seconds PULSE)
• Loss of Unit +24V DC power

When a coal elevation is in service, the associated fuel air dampers modulate as a function of feeder speed as indicated by the associated coal elevation “MODULATING” light coming on.

When the coal elevation is shut down, the associated fuel air dampers are closed, as indicated by a “CLOSED” light .

When the “Post Trip Purge Time Expired” command is established and all FD and ID fans are not off and a “Master fuel trip” command does not exist, the “Open upper fuel air dampers (Elevation H, G, F & E)” command is removed from memory.

Thirty seconds later, the “Open lower fuel air dampers (Elevation D, C, B & A)” command is removed from memory.